Exp No:1.a Analyze the trend of data science job postings over the last Decade

Aim:

To Analyze the trend of data science job postings over the last Decade

Description:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {

'Year':list(range(2016,2027)),

'Job Posting':[200,250,300,380,450,550,660,750,800,880,1000]

}

df = pd.DataFrame(data)

plt.plot(df['Year'],df['Job Posting'],marker='o')

plt.xlabel("Year")

plt.ylabel("Number of Job Posting")

plt.title("Trend of Data Science")

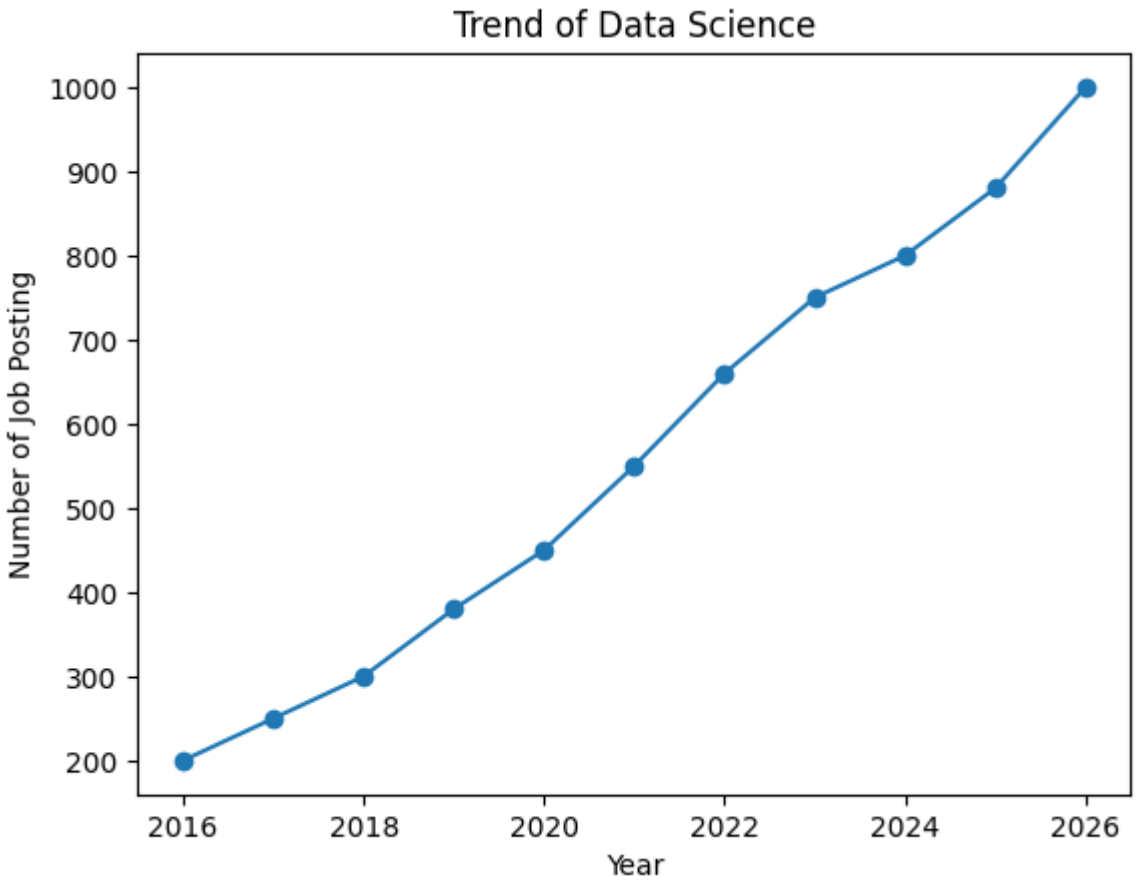
plt.show()
```

Sample Data Input:

Year = 2016,2027

Job Posting = 200,250,300,380,450,550,660,750,800,880,1000

Sample Output:



Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Aim:

To Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

Code:

```
import matplotlib.pyplot as plt

roles=['Data Analyst','Data Engineer','Data Scientist','ML Engineer','Business Analyst']

count = [100,160,300,200,180]

plt.bar(roles,count)

plt.xlabel("Role")

plt.ylabel("Count")

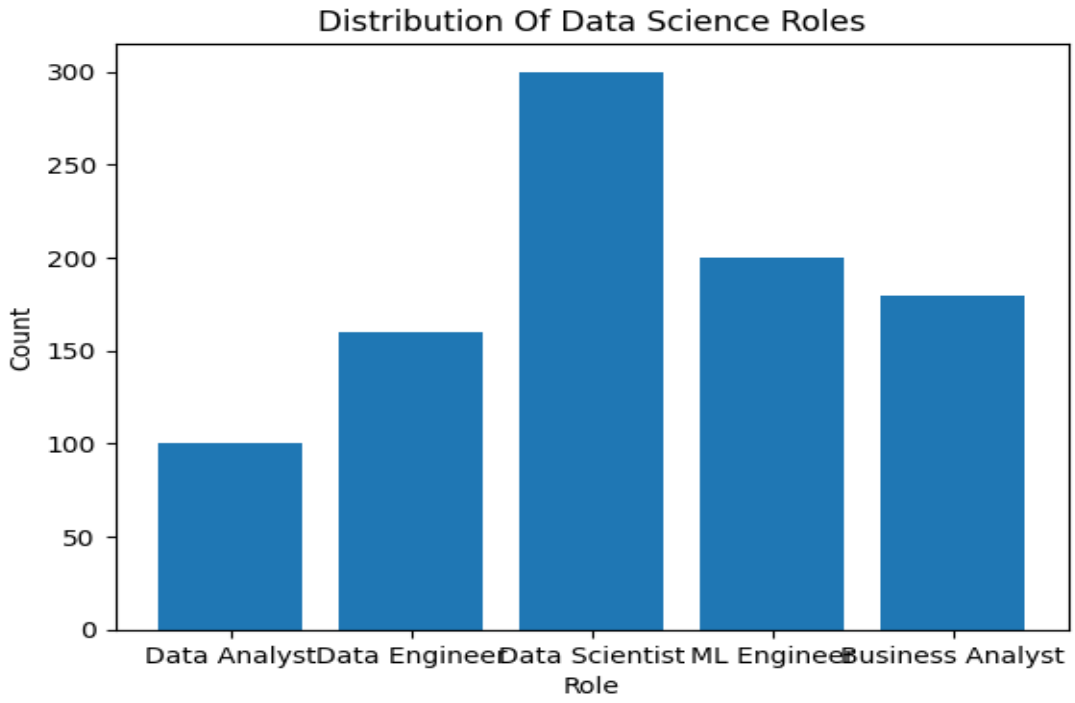
plt.title("Distribution Of Data Science Roles")

plt.show()
```

Sample Data Input:

roles=Data Analyst,Data Engineer,Data Scientist,ML Engineer,Business Analyst

count = 100,160,300,200,180



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Aim:

To Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

Code:

```
import pandas as pd
data = {
  'ID':[1,2,3],
  'Name':['A','B','C'],
  'Age':[18,19,20]
}
structured_data=pd.DataFrame(data)
print('Structured Data:\n',structured_data)
unstructured_data="This is an example of Unstructured data.It can be a piece of text,an image,or a video file."
print('Unstructured Data:\n',unstructured_data)
semi_structured_data = {'ID':1,'Name':'D','Attributes':{'Height':185,'Weight':67}}
print('Semistructured:\n',semi_structured_data)
```

Output:

```
Structured Data:
   ID Name  Age
0  1    A   18
1  2    B   19
2  3    C   20
Unstructured Data:
 This is an example of Unstructured data.It can be a piece of text,an image,or a video file.
Semistructured:
 {'ID': 1, 'Name': 'D', 'Attributes': {'Height': 185, 'Weight': 67}}
```

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Aim:

To Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

encrypt=f.encrypt(b"This Is 1st Experiment")

print('Encrypted : ',encrypt)

decrypt=f.decrypt(encrypt)

print('Decrypted : ',decrypt)
```

Output:

```
Encrypted :
b'gAAAAABqfbufPxHUArBHklsjI1aUfpMZIJYObCPxvWUenWbYToUlDHs7cY3465jfCEpPx-
DLkVFMtRWyZ38nWo4re5kvqomxlbGtS0gGrz_0U6gqgL79mwQ='

Decrypted :  b'This Is 1st Experiment'
```